Table 1: Clustering quality results for the CENSUS1990 dataset.

				Lloyd		Cost			$\text{Gap}^{\text{SFC}}_{\text{Cost}}$ [%]		Balance			
k	Target	p	q	Cost	Balance	S-MPFC	Flow	SFC	S-MPFC	Flow	Lloyd	S-MPFC	Flow	SFC
2	0.90	9	10	3,205,963	0.93	3,234,760	**3,206,079**	3,324,154	-2.69	-3.55	✓	✓	✓	✓
2	0.75	3	4			3,234,760	**3,206,079**	3,236,939	-0.07	-0.95	✓	✓	✓	✓
2	0.50	1	2			3,234,760	**3,206,080**	3,213,830	0.65	-0.24	✓	✓	✓	✓
5	0.90	9	10	2,082,991	0.49	2,436,024	**2,229,905**	2,594,286	-6.10	-14.05	✗	✓	✓	✓
5	0.75	3	4			2,267,625	**2,142,045**	2,464,672	-7.99	-13.09	✗	✓	✓	✓
5	0.50	1	2			2,122,421	**2,083,342**	2,388,713	-11.15	-12.78	✗	✓	✓	✓
10	0.90	9	10	1,619,760	0.46	2,038,966	**1,801,850**	2,218,436	-8.09	-18.78	✗	✓	✓	✓
10	0.75	3	4			1,882,197	**1,712,834**	2,055,310	-8.42	-16.66	✗	✓	✓	✓
10	0.50	1	2			1,717,188	**1,639,519**	1,900,991	-9.67	-13.75	✗	✓	✓	✓
15	0.90	9	10	1,394,466	0.36	1,902,220	**1,613,071**	2,027,372	-6.17	-20.44	✗	✓	✓	✓
15	0.75	3	4			1,740,733	**1,524,018**	1,885,811	-7.69	-19.19	✗	✓	✓	✓
15	0.50	1	2			1,523,456	**1,435,454**	1,758,309	-13.36	-18.36	✗	✓	✓	✓
Average									-6.73	-12.65				

(**bold**) best result; (underline) 2nd best result; (✓) target balance attained; (✗) target balance not attained

Table 2: Running times for the CENSUS1990 dataset.

				CPU_{tot} [s]			CPU_{avg} [s]			$\text{Gap}^{\text{SFC}}_{\text{CPU}_{\text{avg}}}$ [%]		Avg. iterations	
k	Target	p	q	S-MPFC	Flow	SFC	S-MPFC†	Flow	SFC	S-MPFC	Flow	S-MPFC	Flow
2	0.90	9	10	14.85	limit	limit	1.89	90.08	851.35	-99.78	-89.42	2.74	3.56
2	0.75	3	4	14.55	limit	limit	1.88	84.13	739.70	-99.75	-88.63	3.31	3.26
2	0.50	1	2	16.24	limit	limit	1.90	84.08	751.03	-99.75	-88.80	3.35	3.24
5	0.90	9	10	44.92	limit	limit	2.19	180.14	856.94	-99.74	-78.98	3.60	3.37
5	0.75	3	4	38.63	limit	limit	2.13	200.31	746.86	-99.71	-73.18	4.04	3.94
5	0.50	1	2	30.40	limit	limit	2.04	302.46	721.01	-99.72	-58.05	4.12	6.64
10	0.90	9	10	107.60	limit	limit	2.82	458.22	850.03	-99.67	-46.09	3.24	3.86
10	0.75	3	4	68.24	limit	limit	2.42	450.19	745.43	-99.68	-39.61	3.53	4.86
10	0.50	1	2	51.96	limit	limit	2.26	453.31	721.92	-99.69	-37.21	3.42	4.57
15	0.90	9	10	166.27	limit	limit	3.40	522.20	846.87	-99.60	-38.34	2.59	4.00
15	0.75	3	4	102.86	limit	limit	2.77	528.96	737.56	-99.62	-28.28	3.18	3.83
15	0.50	1	2	66.71	limit	limit	2.41	739.10	719.03	-99.66	2.79	2.80	5.75
Average										-99.70	-55.32	3.33	4.24

† including preprocessing time; (limit) time limit of 3,600 seconds reached

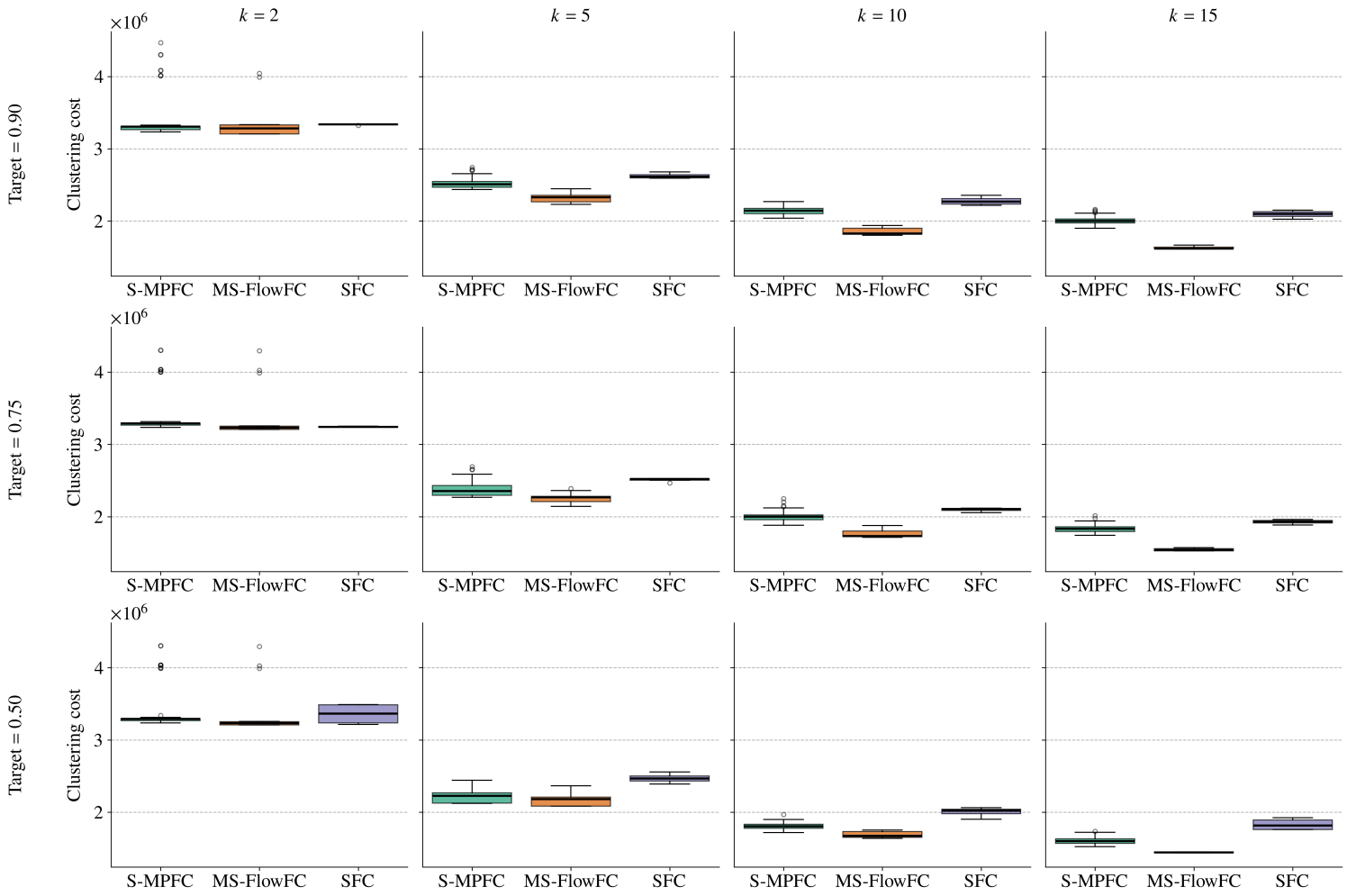

Figure 1: Boxplots of clustering costs for solutions obtained with different random seeds on the CENSUS1990 dataset, across different methods, numbers of clusters, and tolerance values.